

Paul Lee

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EDUCATION

The University of Texas at Austin | Austin, TX

Expected May 2028

Bachelor of Science, Mechanical Engineering | GPA 3.82

- Certificate in Programming and Computation
- Relevant Coursework: Thermodynamics, Dynamics, Statics, Engineering Computing Methods, Software Design, Database

TECHNICAL SKILLS

- **Software:** Solidworks, Creo, Fusion 360, ANSYS, Onshape, COMSOL, Python, MATLAB, PLM, EZ-toI, SQL, Aladdin, Excel
- **Technical:** CAD, FEA, 2D Drawing, 3D Printing, DFMA, Tolerance Analysis, CFD, Woodworking, GD&T, Soldering
- **Languages:** English (Native), Korean (Native)
- **Certificates:** CSWA, CornellIX(FEA/CFD), CAM and Design Manufacturing(Fusion 360)

ENGINEERING EXPERIENCE

Celestica Mechanical Engineering Intern | Dallas, TX

May 2026 - August 2026

- Redesigned a sheet metal fan tray with a custom sliding mechanism, saving ~3.5 metric tons of steel (~\$10K in raw material cost) across a 60K-unit production run
- Designed sheet metal chassis, PCBA component placements, and power cable routing in Creo, reducing assembly time by up to 50% across AI server assemblies
- Revised 150+ part and assembly drawings, applying GD&T and ballooning inspection-critical dimensions
- Supported validation of blades through thermal, leak, seismic, tilt, and roll testing to verify product performance
- Performed worst-case and RSS tolerance stackup analysis and global interference checks across large TLA assemblies

Texas Aerial Robotics Hardware Engineer | Austin, TX

Aug 2026 - Present

- Designed aluminum tube chassis for a 200 lb UGV; selected material and profile size from load and weight requirements
- Performed FEA on various components, including 6061-T6 steering plate to evaluate stress, deformation, and yielding under tie-rod loads
- Modeled an Ackermann steering geometry in Onshape, including wishbones, tie rods, and rack and pinion assembly

DUCK Lab Undergraduate Research Assistant | Austin, TX

Oct 2024 - May 2025

- Increased annual energy yield by 8.2% and projected profit by 40.7% through solar farm layout optimization
- Created 50+ solar farm layouts in Aladdin while cutting design generation time by 11x

MINIMAX Lab Undergraduate Research Assistant | Austin, TX

Sept 2025 - Dec 2025

- Modeled a 125 μm silicon AFM cantilever in COMSOL, extracting stress distribution, spring constant, and the first three flexural modes for probe calibration
- Reviewed literature on active AFM cantilevers and piezoelectric actuation to inform model parameters

PROJECTS

Hero Robot

- Redesigned the chassis belly pan and box tube, redimensioning all misaligned mounting holes across the assembly
- Designed and prototyped rounded skirt geometries that reduced collision forces by ~40%

Crates and 1V1 Arena

- Designed crates & 50 ft $\times$ 50 ft PVC arena from using McMaster-Carr components, reducing testing setup time by ~30%
- Fabricated wooden barriers for competition using miter saws, planers, and drills

Robotathon

- Developed an autonomous robot capable of color detection, line following, and obstacle navigation
- Fabricated custom chassis components using 3D printing, including battery and IR sensor mounts, and ball scooper

HONORS & AWARDS

- University Honors (Fall 2024, Spring 2025)
- 1st place, Robotathon 2024